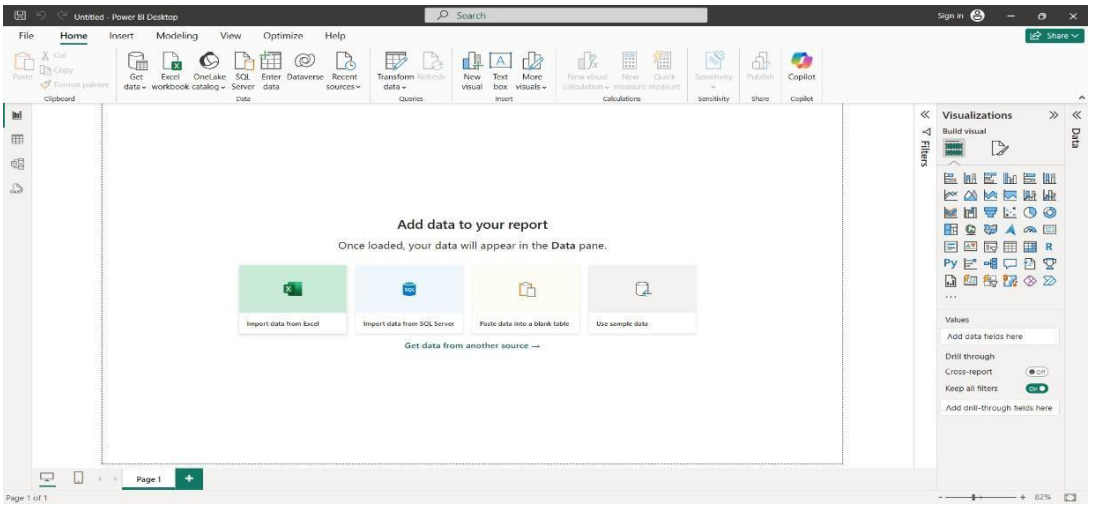
EX.NO: 8	K-MEANS CLUSTERING USING R PROGRAMMING
DATE:	

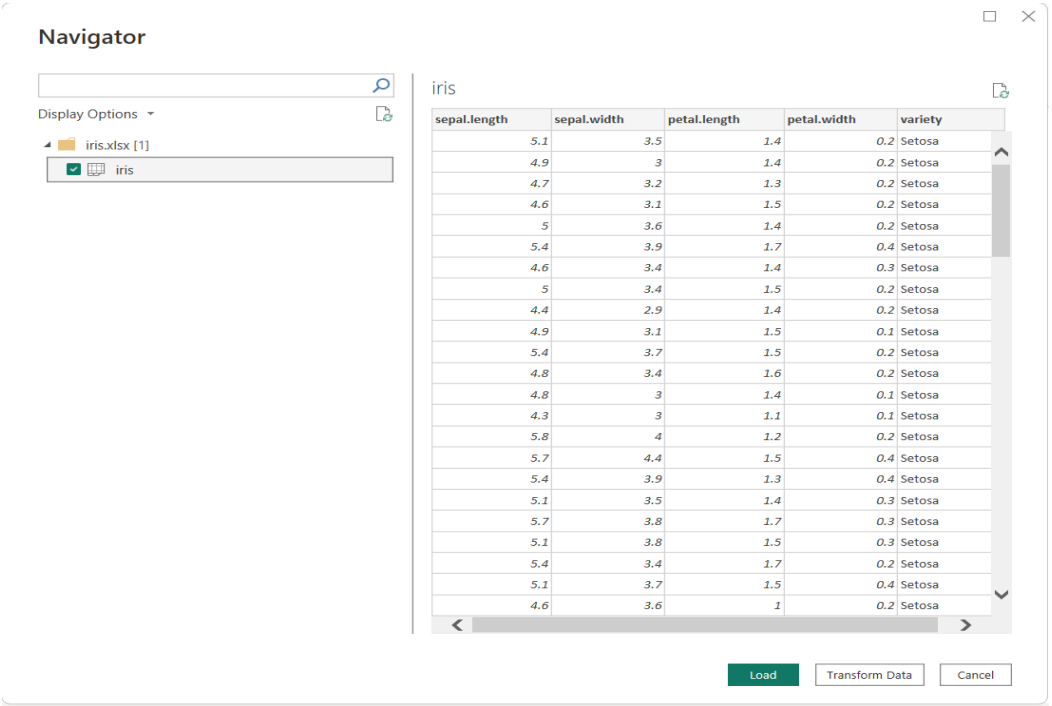
AIM:

To implement the K-Means Clustering using R programming in Power BI.

PROCEDURE:

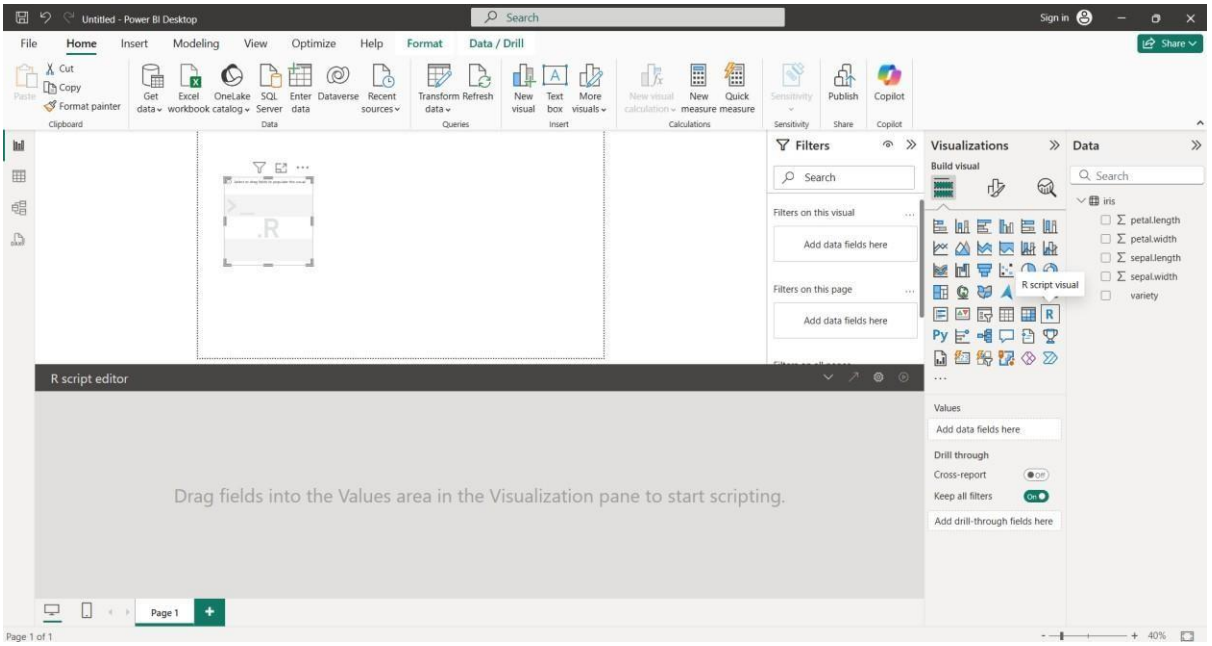
Step 1: Import the Dataset into Power BI



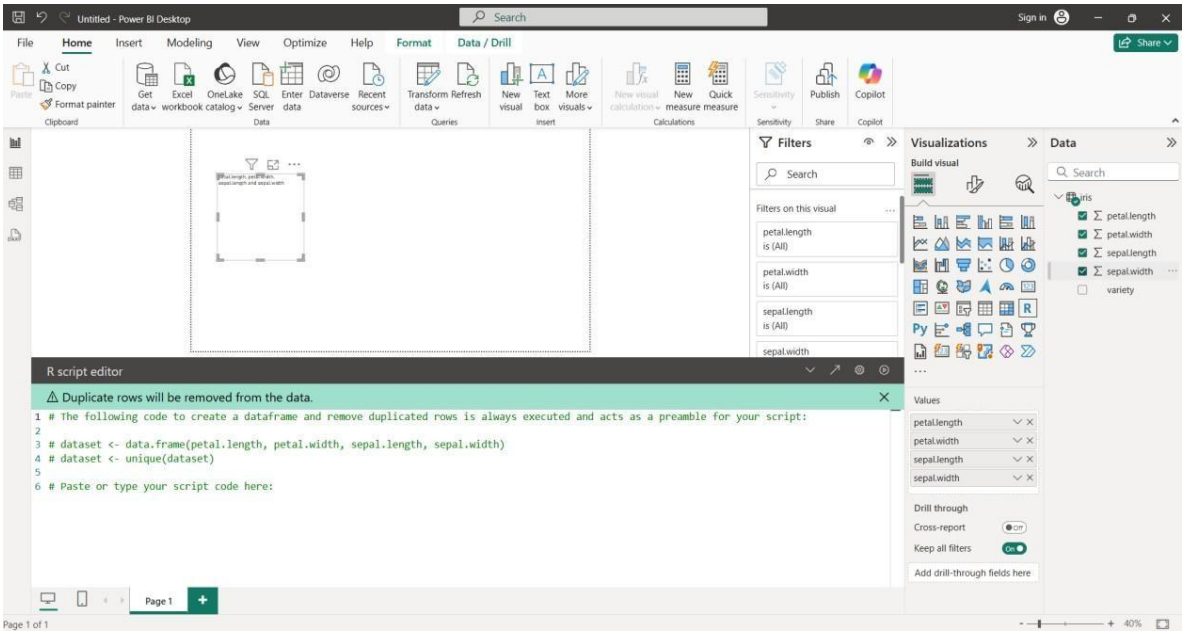


sepal.length	sepal.width	petal.length	petal.width	variety
5.1	3.5	1.4	0.2	Setosa
4.9	3	1.4	0.2	Setosa
4.7	3.2	1.3	0.2	Setosa
4.6	3.1	1.5	0.2	Setosa
5	3.6	1.4	0.2	Setosa
5.4	3.9	1.7	0.4	Setosa
4.6	3.4	1.4	0.3	Setosa
5	3.4	1.5	0.2	Setosa
4.4	2.9	1.4	0.2	Setosa
4.9	3.1	1.5	0.1	Setosa
5.4	3.7	1.5	0.2	Setosa
4.8	3.4	1.6	0.2	Setosa
4.8	3	1.4	0.1	Setosa
4.3	3	1.1	0.1	Setosa
5.8	4	1.2	0.2	Setosa
5.7	4.4	1.5	0.4	Setosa
5.4	3.9	1.3	0.4	Setosa
5.1	3.5	1.4	0.3	Setosa
5.7	3.8	1.7	0.3	Setosa
5.1	3.8	1.5	0.3	Setosa
5.4	3.4	1.7	0.2	Setosa
5.1	3.7	1.5	0.4	Setosa
4.6	3.6	1	0.2	Setosa

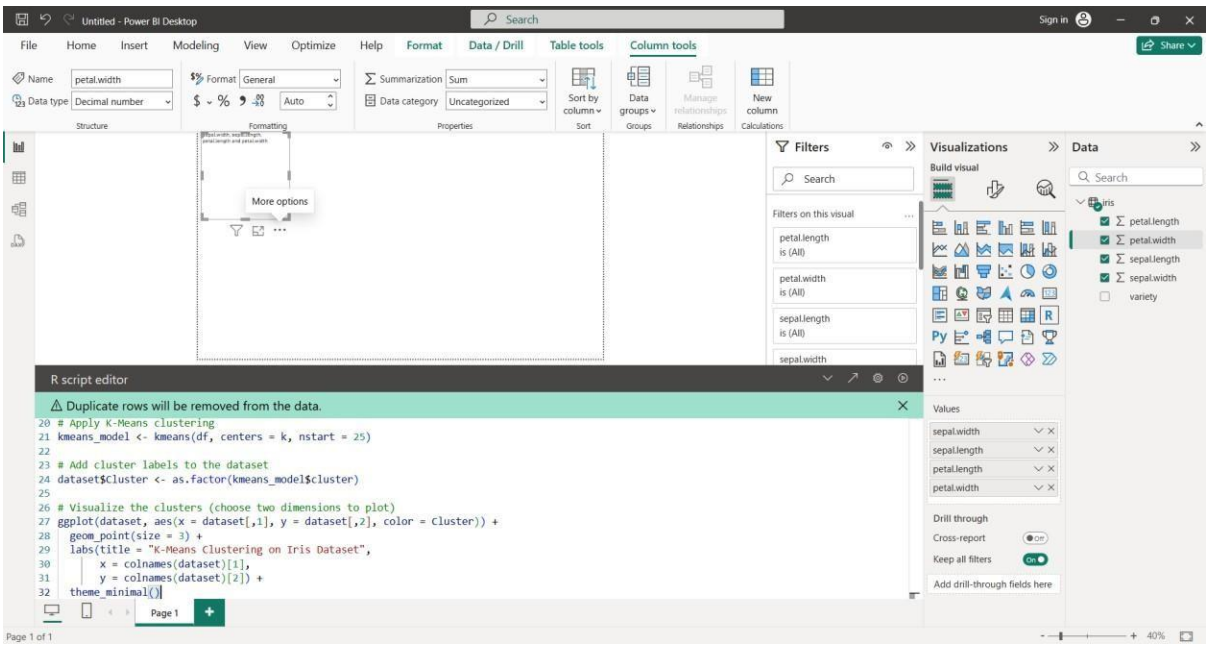
Step 2: Then select R script visual from the Visualization



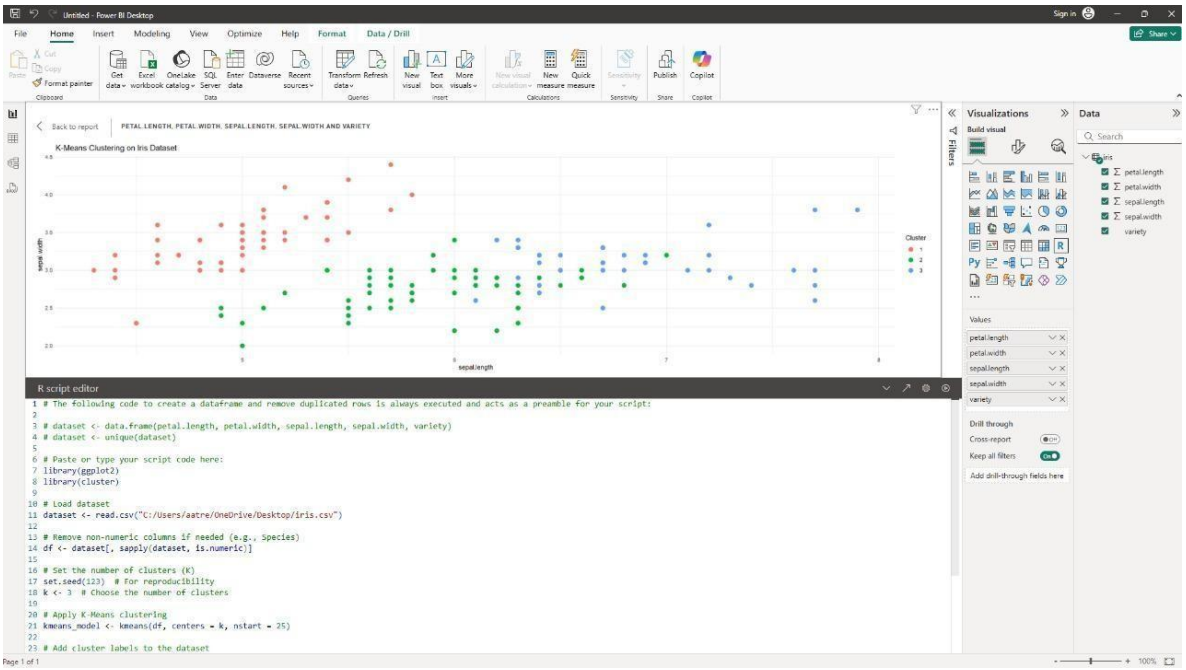
Step 3: Select No of Parameters from the dataset to work within it.



Step 4: In the Script editor Type the K-Means algorithm using R for visualization



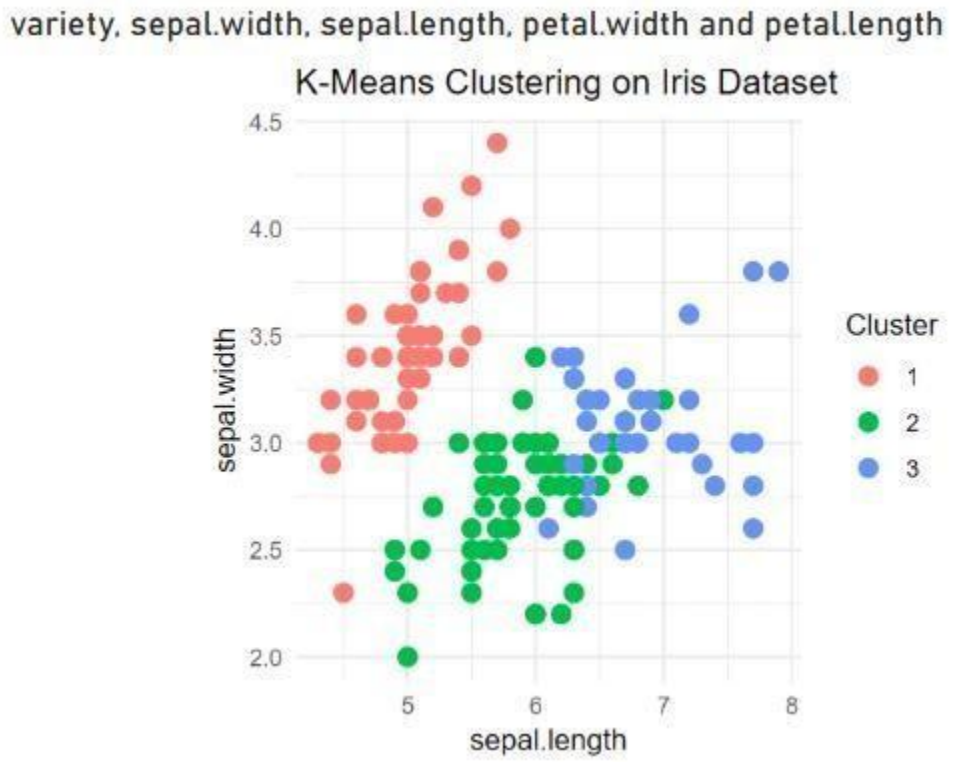
Step 5: Run the script for the visualization of the code



CODE:

```
library(ggplot2)
library(cluster)
# Load dataset
dataset <- read.csv("C:/Users/AI_LAB/Desktop/iris.csv")
# Remove non-numeric columns if needed (e.g., Species)
df <- dataset[, sapply(dataset, is.numeric)]
# Set the number of clusters (K)
set.seed(123)  # For reproducibility
k <- 3  # Choose the number of clusters
# Apply K-Means clustering
kmeans_model <- kmeans(df, centers = k, nstart = 25)
# Add cluster labels to the dataset
dataset$Cluster <- as.factor(kmeans_model$cluster)
# Visualize the clusters (choose two dimensions to plot)
ggplot(dataset, aes(x = dataset[,1], y = dataset[,2], color = Cluster)) +
geom_point(size = 3) +
  labs(title = "K-Means Clustering on Iris Dataset",
       x = colnames(dataset)[1],
       y = colnames(dataset)[2]) +
  theme_minimal()
```

OUTPUT:



RESULT:

Thus, the implementation of K-Means Clustering using R programming in Power BI was executed successfully and the output also verified.